COMPUTATIONAL BEAM ANALYSIS

Python • PySide6 • NumPy • Matplotlib • JSON

Overview

A Python-based structural engineering application for analytical beam analysis. Enter beam parameters through the graphical interface, calculate structural responses, and visualize deflection, shear force, and bending moment.

Key Features

- **Graphical user interface:** Interactive desktop application built with PySide6.
- **Analytical beam solver:** Calculates support reactions, maximum shear force, bending moment, and deflection for the implemented beam cases.
- **Response visualization:** Plots the deflected shape, Shear Force Diagram (SFD), and Bending Moment Diagram (BMD).
- **JSON input save/load:** Save beam inputs to JSON files and reload previously saved configurations.
- **Input validation:** Checks numerical inputs and supported loading configurations.
- **Calculation validation:** Representative cases are compared with independent manual calculations.

Supported Beam Configurations

Boundary condition	Loading condition
Simply Supported	Point Load
Simply Supported	Full-span Uniformly Distributed Load (UDL)
Cantilever	Point Load
Cantilever	Full-span Uniformly Distributed Load (UDL)

Limitations: Fixed-Fixed and Propped Cantilever beams are not supported unless their solvers are implemented and verified.

Analytical Formulas

The application uses Euler–Bernoulli beam theory for linear-elastic static analysis. Assumptions include constant flexural rigidity EI, small deflections and rotations, idealized supports, and neglected shear deformation.

1. Simply Supported Beam — Point Load

For point load P at distance a from the left support; b = L – a:

$$R_A = Pb/L \quad R_B = Pa/L$$

$$V_{max} = \max(R_A, R_B) \quad M_{max} = Pab/L$$

Let c = min(a, b):

$$\delta_{max} = Pc(L^2 - c^2)^{3/2} / (9\sqrt{3}\,LEI)$$

For a central point load (a = b = L/2): $M_{max} = PL/4$; $\delta_{max} = PL^3/(48EI)$.

2. Simply Supported Beam — Full-Span UDL

$$R_A = R_B = wL/2$$

$$V_{max} = wL/2 \quad M_{max} = wL^2/8 \text{ (at midspan)}$$

$$\delta_{max} = 5wL\blacksquare/(384EI) \text{ (at midspan)}$$

3. Cantilever Beam — Point Load

For point load P at distance a from the fixed end, 0 < a ≤ L:

$$R_A = P \quad |M_A| = Pa$$

$$V_{max} = P \quad M_{max} = Pa$$

$\delta_{max} = Pa^2(3L - a)/(6EI)$ **(free end)**

For a free-end load (a = L): $\delta_{max} = PL^3/(3EI)$.

4. Cantilever Beam — Full-Span UDL

$R_A = wL$ $|M_A| = wL^2/2$

$V_{max} = wL$ $M_{max} = wL^2/2$

$\delta_{max} = wL\blacksquare/(8EI)$ **(at free end)**

Units

Parameter	Unit	Parameter	Unit
Beam length L	m	Point load P	kN
UDL intensity w	kN/m	Point-load position a	m
Young's modulus E	MPa (N/mm²)	Second moment I	mm■

Outputs: reactions and shear in kN; moment in kN·m; deflection in mm; plot positions in m. Internally, the solver uses N–mm units. Numerically, 1 kN/m = 1 N/mm.

Project Files

```
main.py — application entry point
GUI.py — graphical interface
Analytical_solver.py — analytical calculations
visualization.py — response plots
I_O_module.py — JSON input save/load
requirements.txt — dependencies
Launch_beam_analysis.bat — Windows launcher
```

Installation and Launch

Requirements: Python installed and available from the terminal. Required packages are listed in requirements.txt.

Option 1 — Launch using the Windows batch file

1. Keep Launch_beam_analysis.bat in the same folder as main.py.
2. Install dependencies once from a terminal in the project folder:

```
python -m pip install -r requirements.txt
```

3. Double-click Launch_beam_analysis.bat to start the application.

Option 2 — Launch using main.py

Open a terminal in the project folder and run:

```
python main.py
```

Validation

- Simply supported beam with a central point load.
- Simply supported beam with a full-span UDL.
- Cantilever beam with a point load at the free end.
- Cantilever beam with a full-span UDL.

Validation covers support reactions, maximum shear force, bending moment, deflection, and expected locations for representative cases. The validation PDF and spreadsheet may be included when approved for public sharing.

Assumptions and Limitations

- Prismatic beam with constant EI and linear-elastic material behaviour.
- Small deflections and rotations; Euler–Bernoulli theory with shear deformation neglected.
- Idealized supports, static downward vertical loading, and full-span UDLs.
- Only implemented and verified beam cases are supported.

- This tool is not a design-code compliance check; independently verify results before real-world use.

Technology

Python • PySide6 • NumPy • Matplotlib • JSON • Git • GitHub

Disclaimer

For educational, analytical, and engineering-support purposes. Users are responsible for verifying the suitability and accuracy of results before real-world application.